EXPERIMENT 4

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AIM: To develop a blockchain-powered web application using Solidity programming language on Remix IDE and embedding Metamask Wallet

THEORY:

1. MetaMask

MetaMask is a **cryptocurrency wallet and browser extension** that allows users to interact with blockchain networks and decentralized applications (DApps). It manages the user's blockchain accounts, including their public wallet address and private keys, and allows users to approve or reject blockchain transactions. In this experiment, MetaMask acts as the user's wallet for connecting to a blockchain test network and signing transactions generated through Remix IDE.

MetaMask can be connected to Remix IDE so that a Solidity smart contract deployed from Remix uses the account selected in MetaMask. When a transaction is initiated, MetaMask displays the transaction details, such as the destination address, gas fee, and network, and asks the user to confirm the transaction.

2. Testnet

A **testnet** is a blockchain network designed for development and testing purposes. It provides an environment that behaves similarly to a real blockchain network but uses **test cryptocurrency**, which has no real monetary value. Developers can use testnets to deploy and test smart contracts and perform transactions without risking real funds.

In this experiment, the **Sepolia Testnet** or **RSK Testnet** can be used. Test ETH on Sepolia or test RBTC on RSK is used to pay transaction fees during testing. The use of a testnet allows the smart contract to be compiled, deployed, and interacted with safely before considering deployment on a production blockchain network.

3. Remix IDE

Remix IDE is a browser-based development environment used for writing, compiling, deploying, and interacting with Ethereum-compatible smart contracts written in Solidity. It provides a Solidity code editor, compiler, deployment interface, and tools for interacting with deployed contracts.

For this experiment, Remix IDE is connected to MetaMask through the **Injected Provider - MetaMask** environment. This allows transactions initiated from Remix to be signed using the selected MetaMask account.

4. Solidity Smart Contracts

A **smart contract** is a program deployed on a blockchain that automatically executes predefined logic when its functions are called. Solidity is a programming language commonly used to write smart contracts for Ethereum-compatible blockchain networks.

In this experiment, a simple Solidity smart contract based on the selected mini-project is created. The contract is first compiled in Remix IDE and then deployed to the selected testnet through the MetaMask wallet.

5. Connecting MetaMask with Remix IDE

The general process for connecting MetaMask with Remix IDE is:

1. Install the MetaMask browser extension.
2. Create a new wallet or import an existing account.
3. Select the required test network, such as **Sepolia**.
4. Obtain test cryptocurrency from a testnet faucet to pay transaction fees. 5.
Open Remix IDE and create the Solidity smart contract.
6. Compile the smart contract using the Solidity Compiler.
7. Open **Deploy & Run Transactions** in Remix.
8. Select **Injected Provider - MetaMask** as the environment.
9. MetaMask will request permission to connect the selected account to Remix. 10.
Approve the connection.
11. Deploy the compiled smart contract.
12. Confirm the transaction through MetaMask.
13. After the transaction is confirmed, interact with the deployed smart contract through Remix.

6. Transactions and Gas

A **transaction** is an operation submitted to the blockchain that changes its state, such as deploying a smart contract or modifying data stored by a smart contract. Transactions must be signed by the user's blockchain account.

Gas represents the computational work required to execute an operation on the blockchain. A transaction requires a certain amount of gas, and the user pays a transaction fee using the native cryptocurrency of the selected network.

When a transaction is initiated from Remix, MetaMask displays the transaction and its associated fee. The user can review the details and confirm the transaction. Once confirmed and included in a block, the transaction receives a transaction hash that can be used to view its status and details on a blockchain explorer.

7. Blockchain Explorer

A **blockchain explorer** is a web-based tool used to inspect blockchain activity. It allows users to

search for and view information about transactions, blocks, wallet addresses, and smart contracts.

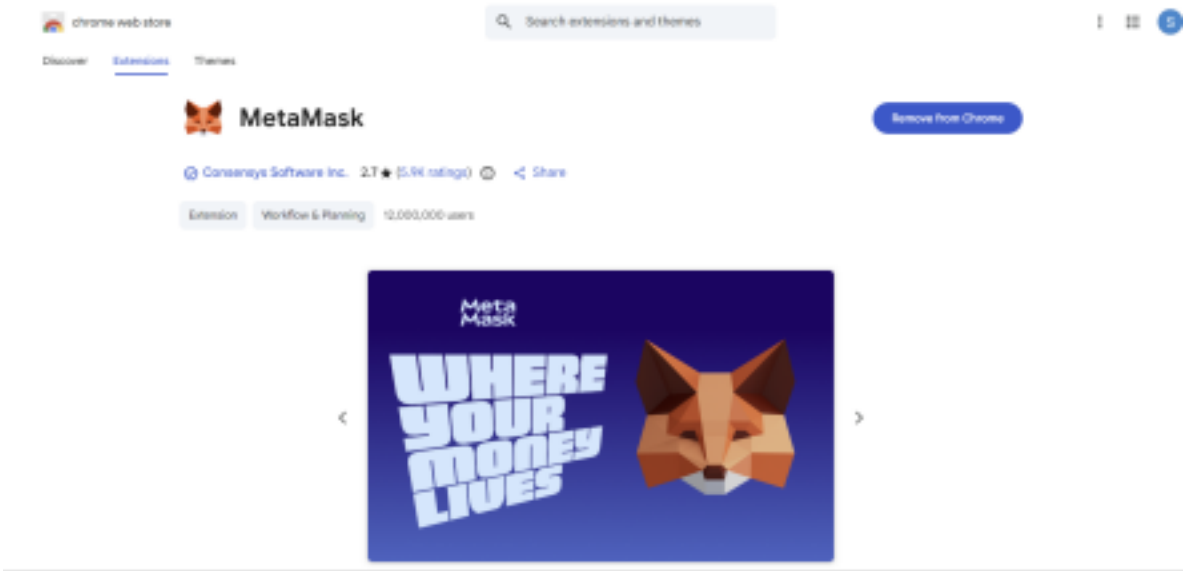
After deploying and interacting with the smart contract, the transaction hash can be viewed on the appropriate testnet explorer. The explorer can show details such as:

- Transaction hash
- Sender address
- Receiver/contract address
- Transaction status
- Block number
- Gas used
- Transaction fee
- Timestamp

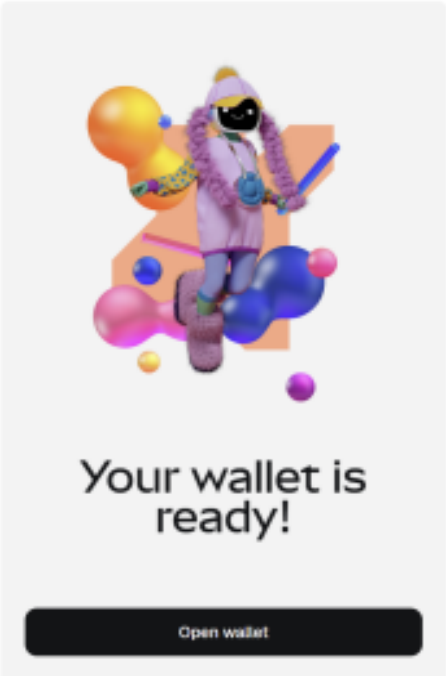
8. Interaction with the Smart Contract

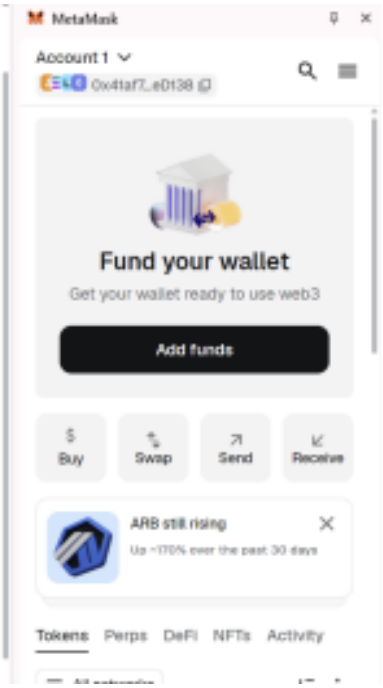
After successful deployment, Remix displays the deployed contract and its available functions. Functions that modify blockchain data require transactions and therefore require confirmation through MetaMask. Read-only functions can generally be called without submitting a state-changing transaction.

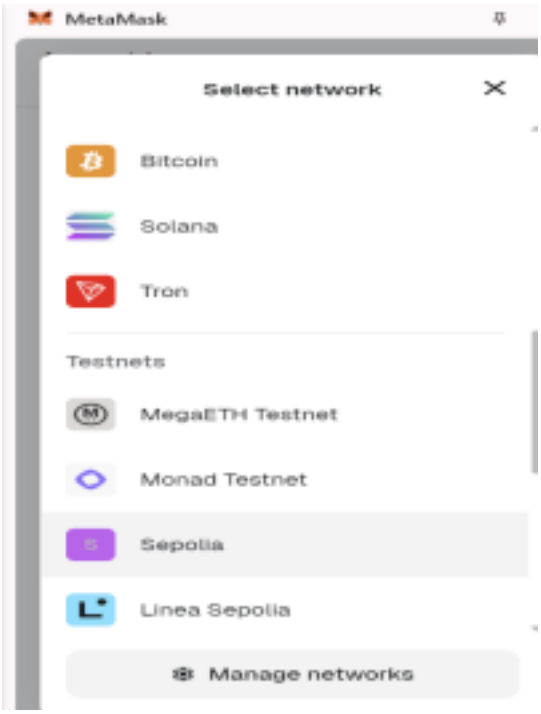
CODE AND OUTPUT:

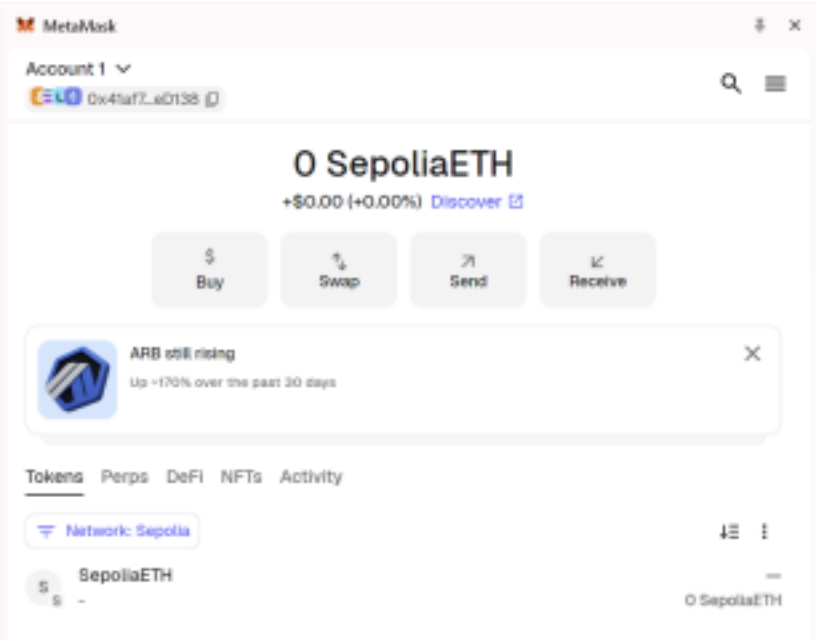



English
MetaMask
Create a new wallet
I have an existing wallet

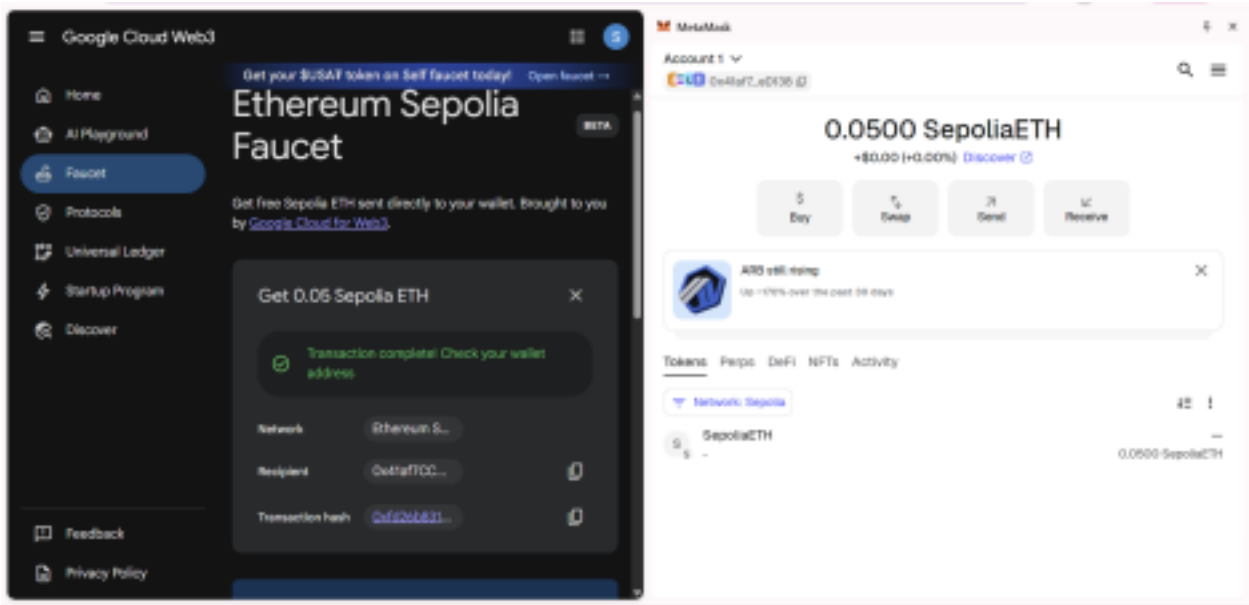

Your wallet is ready!
Open wallet


MetaMask
Account 1
Fund your wallet
Get your wallet ready to use web3
Add funds
Buy
Swap
Send
Receive
ARB still rising
Tokens
Perps
DeFi
NFTs
Activity

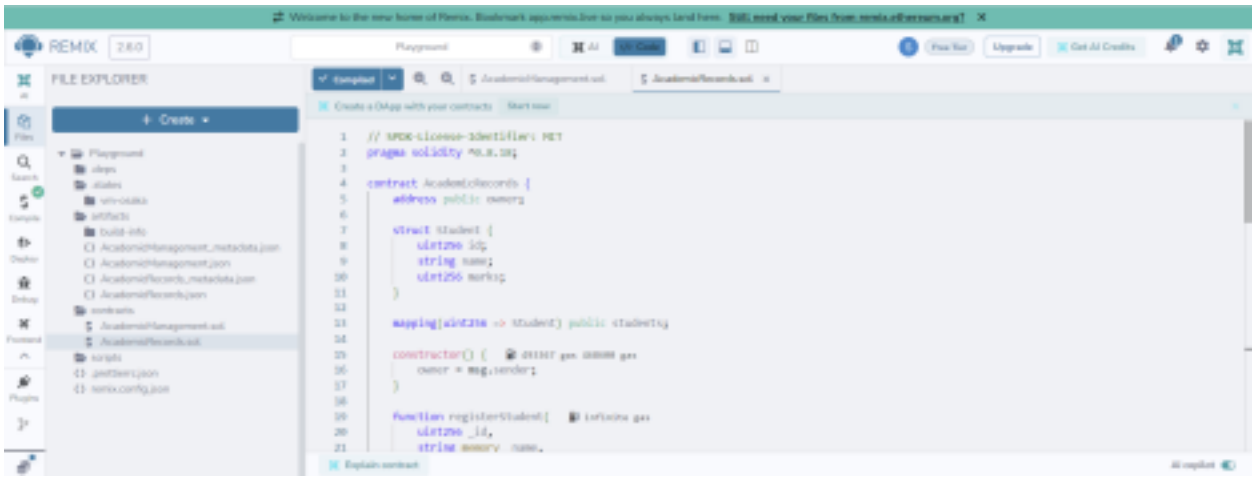




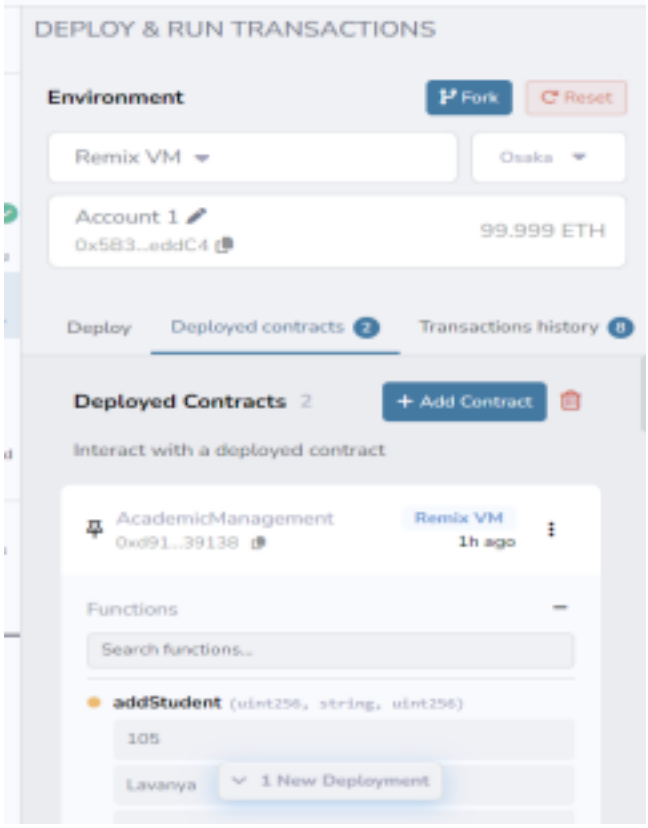
GOOGLE CLOUD FAUCET:



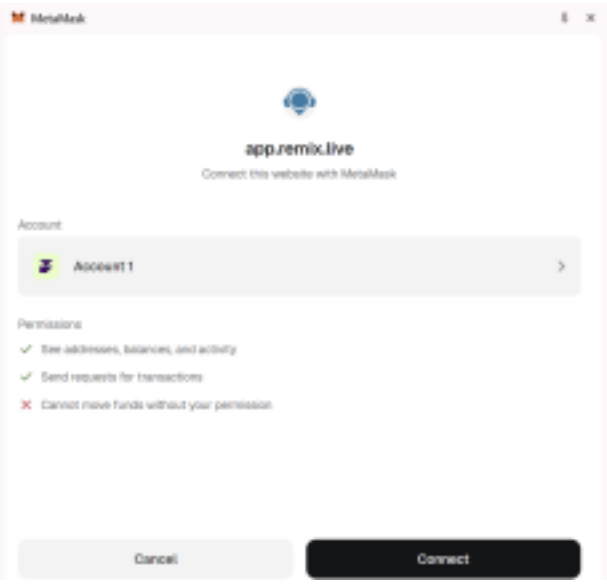
Academic Record Management code:

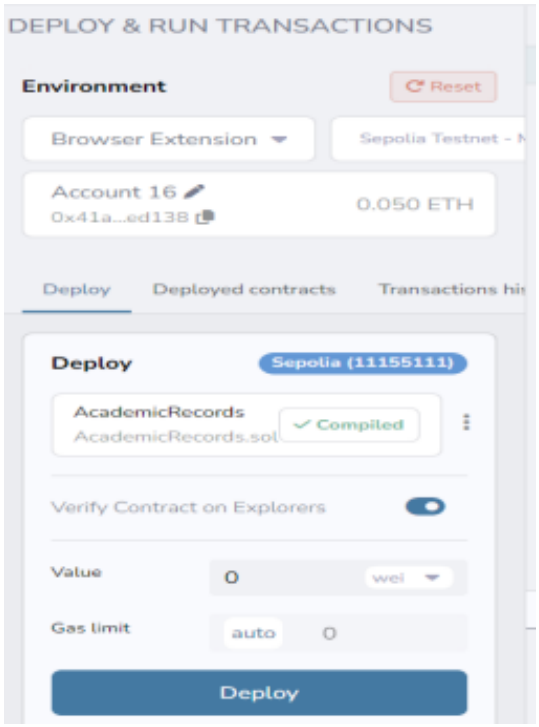


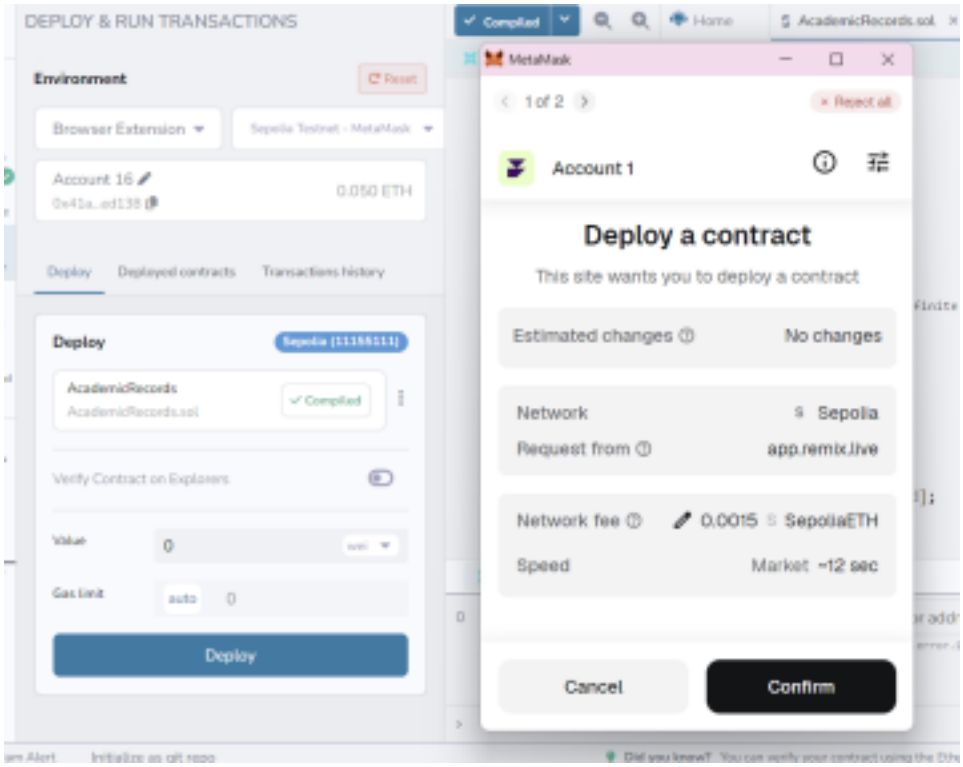
Deploying using MetaMask:

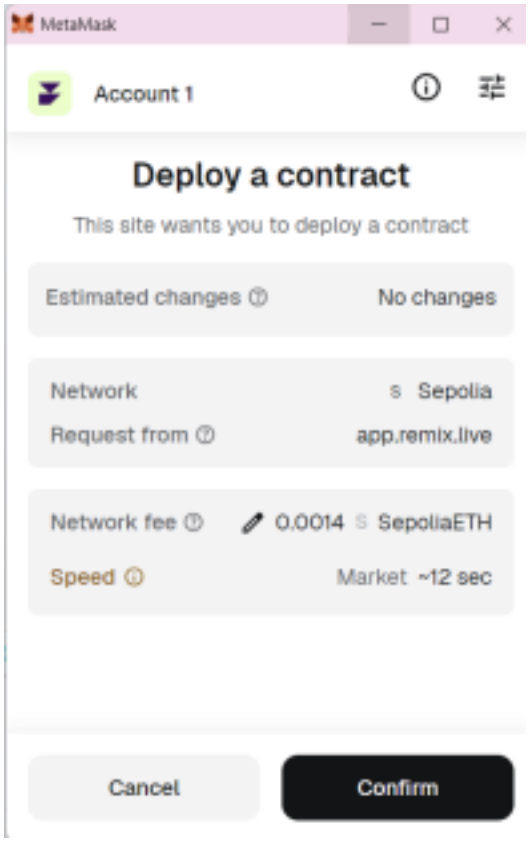

DEPLOY & RUN TRANSACTIONS
Environment
Fork
Reset
Remix VM
Osaka
Account 1
0x5B3...eddC4
99.999 ETH
Deploy
Deployed contracts 2
Transactions history 8
Deployed Contracts 2
+ Add Contract
Interact with a deployed contract
AcademicManagement
0xd91...39138
Remix VM
1h ago
Functions
Search functions...
addStudent (uint256, string, uint256)
105
Lavanya
1 New Deployment

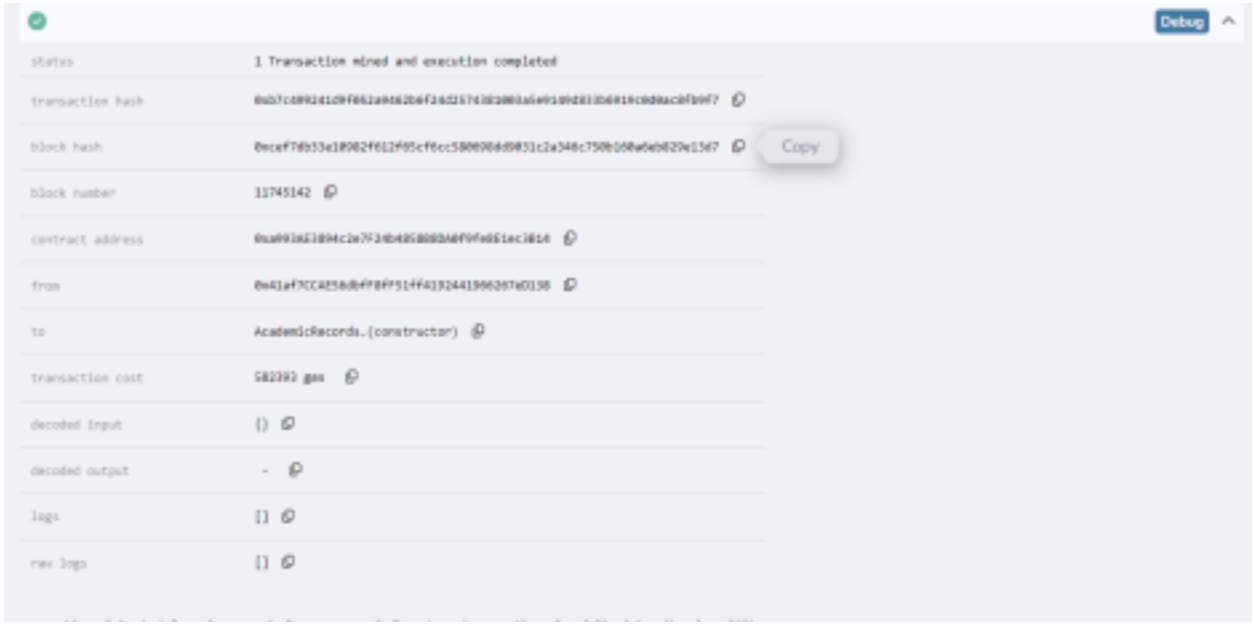

DEPLOY & RUN TRANSACTIONS
Environment
Deployed Contracts
+ Add Contract

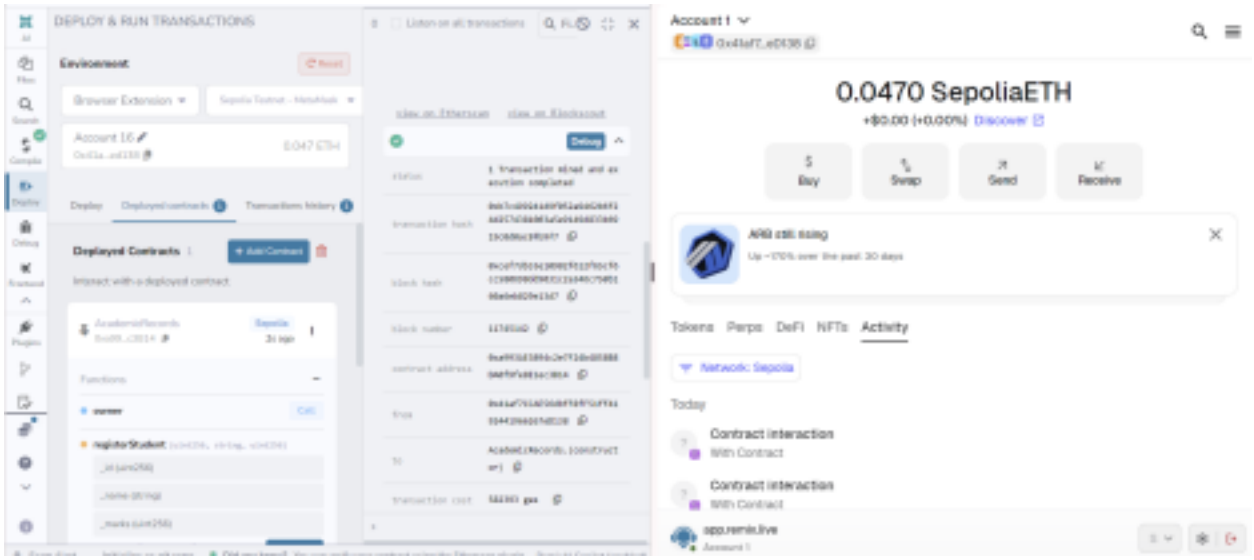

MetaMask
app.remix.live
Connect this website with MetaMask
Account
Account 1
Permissions
Cancel
Connect

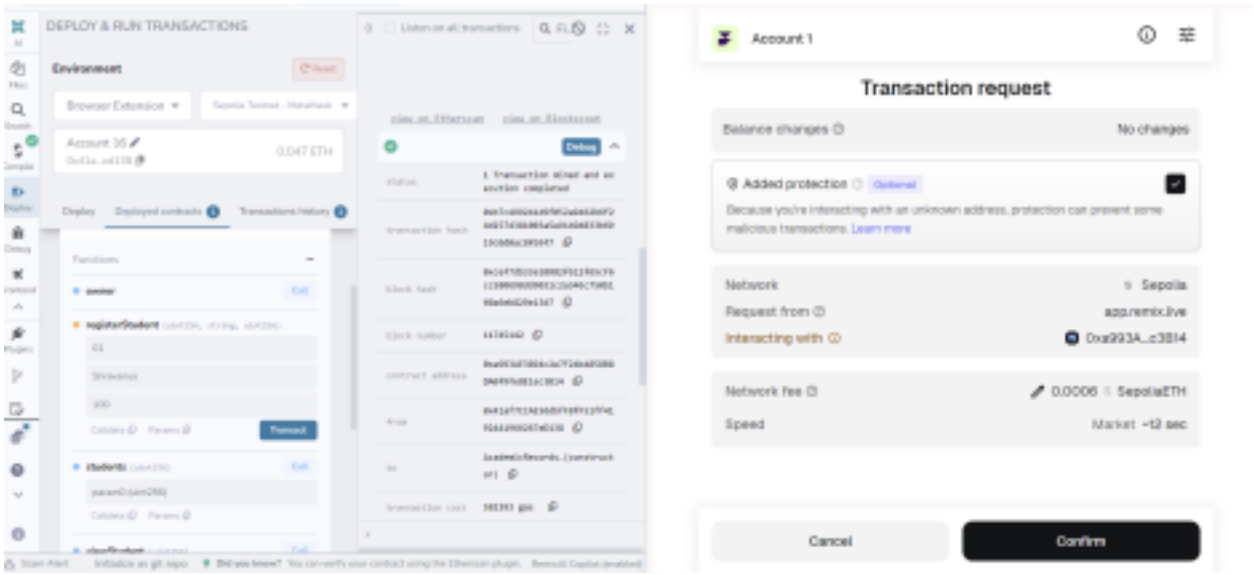

DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
Account 16
0x41a...ed138
0.050 ETH
Deploy
Deployed contracts
Deploy
Sepolia (11155111)
AcademicRecords
AcademicRecords.sol
Compiled
Verify Contract on Explorers
Value
0
wei
Gas limit
auto
0
Deploy


DEPLOY & RUN TRANSACTIONS
Environment
Browser Extension
Account 16
0.050 ETH
Deploy
Deployed contracts
Transactions history
Deploy
AcademicRecords
Compiled
Verify Contract on Explorers
Value
0
Gas limit
auto
0
Deploy
MetaMask
1 of 2
Reject all
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
app.remix.live
Network fee
0.0015
SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm


MetaMask
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
app.remix.live
Network fee
0.0014 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm


Debug
status
1 Transaction mined and execution completed
transaction hash
block hash
Copy
block number
11745142
contract address
from
to
AcademicRecords.(constructor)
transaction cost
582393 gas
decoded input
{}
decoded output
-
logs
[]
raw logs
[]





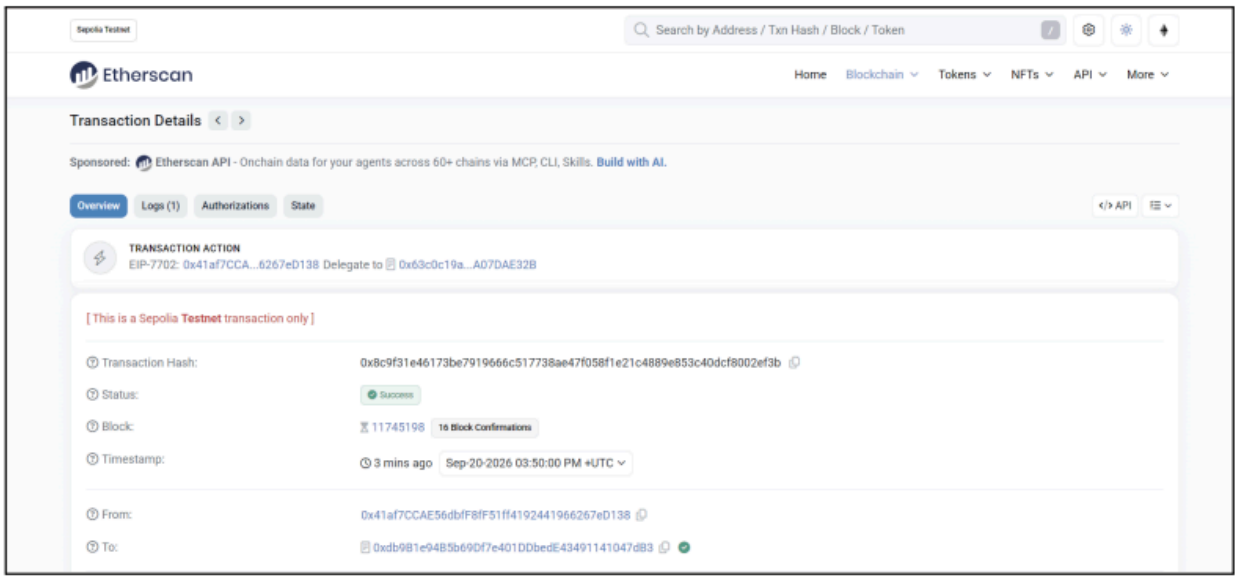
Register student:



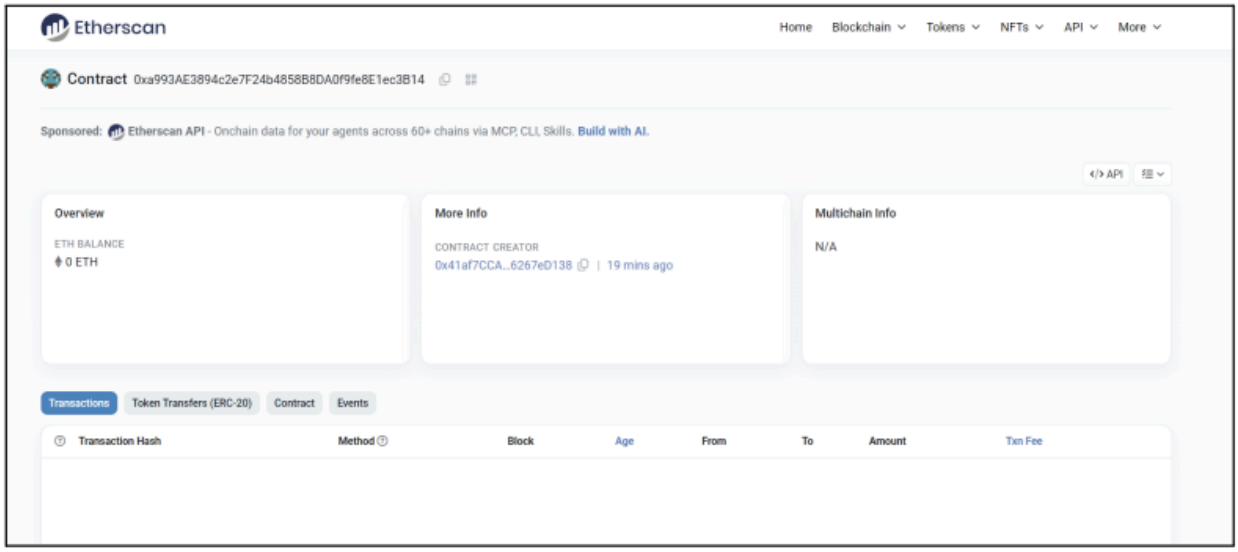
View Student:



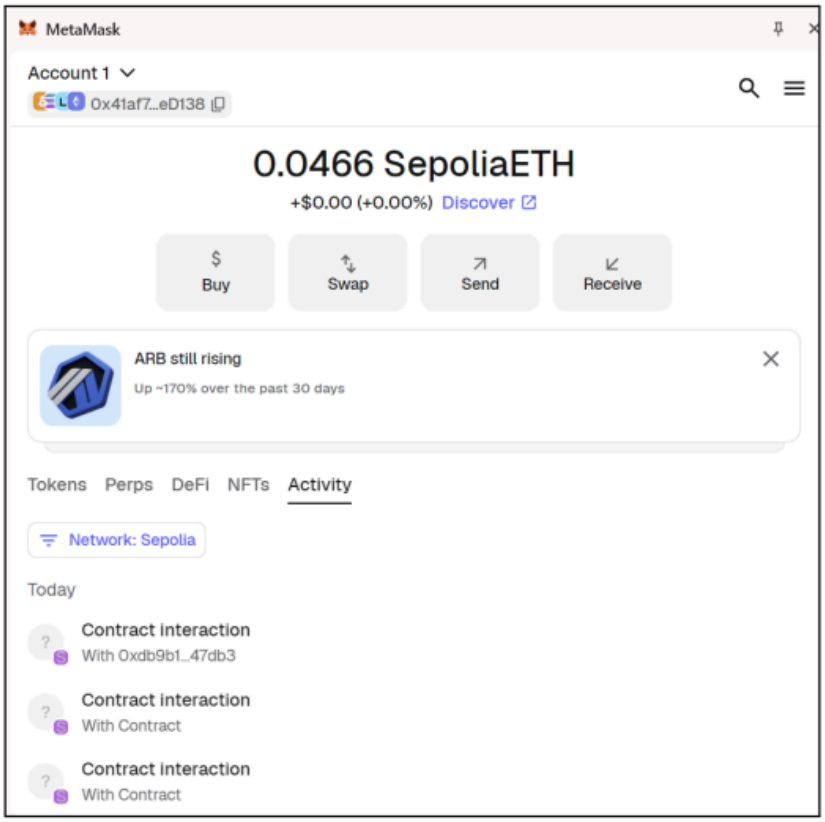
Registration hash verified:



Deployed Contract Address verified:



METAMASK Balance:



CONCLUSION:

The experiment successfully demonstrated the development and deployment of a Solidity-based blockchain application using Remix IDE and MetaMask. The `AcademicRecords` smart contract was compiled and deployed on the Sepolia Testnet, and student information was successfully stored and retrieved through blockchain transactions. MetaMask was used to approve and execute transactions, while Sepolia Etherscan was used to verify the transaction details. The experiment provided practical understanding of smart contract deployment, wallet integration, blockchain transactions, gas consumption and interaction with deployed contracts.